

AJAY BABU NANNAPANENI

Software Engineer Intern | Full Stack Development | AI & Data-Driven Applications
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EDUCATION

University of South Florida, Tampa, FL <i>Master of Science in Computer Science</i>	Aug 2023 - May 2025 GPA 3.5/4
KKR & KSR INSTITUTE OF TECHNOLOGY AND SCIENCES, Guntur, India <i>Bachelor of Technology in Computer Science and Engineering</i>	July 2017 - May 2021 GPA 3.66/4

TECHNICAL SKILLS

Programming & Frameworks: C#, .NET, ASP.NET Core, Python, Java, Go, React.js, Node.js, TypeScript, JavaScript, HTML5, CSS3, Unity 2D.

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Terraform, Git, GitHub Actions, Jenkins, CI/CD pipelines.

AI & ML Technologies: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Power BI, Data Visualization, Predictive Modeling, ML Pipelines, LLM Integration.

Software Development Methodologies: Object-Oriented Design, Design Patterns, RESTful APIs, Microservices, Agile/Scrum, Test-Driven Development, Code Reviews, Version Control.

EXPERIENCE

Software Engineer Intern – AI & ML , Infrastructure Consulting and Engineering, Tampa ,FL • Automated crash classification on 500+ datasets, reducing manual effort by 60% and accelerating safety reporting. • Improved data accuracy by 35% using anomaly detection pipelines, ensuring reliable insights for decision-makers. • Developed AI-driven pipelines on AWS, cutting operational costs and enabling scalable adoption across teams. • Collaborated across the full SDLC (design, development, testing, deployment), ensuring high-quality, maintainable software solutions.	Jan 2025 - Present
Graduate Research Assistant, CUTR , University of South Florida, Tampa ,FL • Analyzed 300+ crash datasets with Python & AI, revealing patterns that advanced safety research. • Built 5+ Power BI dashboards to visualize crash trends, driving data-informed policy decisions. • Developed backend modules for scheduling and storage, boosting system scalability. • Optimized data workflows, cutting processing time by 40% and accelerating analytics delivery.	April 2024 - May 2025
Associate software Engineer, Light and Wonder, Bangalore, India • Delivered 7+ C2/C3 casino games in Unity & C#, meeting compliance standards for WAL, NYL, TITAN, OLG & SA markets. • Built and optimized server-client messaging, ensuring 100% reliable state synchronization with proper State Machines. • Resolved 50+ critical bugs on Titan market games, ensuring stability and timely release. • Enhanced Unity performance, cutting load times by 30% and improving frame stability.	Oct 2021 - July 2023

CERTIFICATIONS

- Programming in Java – NPTEL (National Programme on Technology Enhanced Learning)
- AWS Fundamentals: Going Cloud-Native – Coursera (Amazon Web Services)
- Python for Data Science, AI & Development – Coursera (IBM)
- Python Data Structures – Coursera (University of Michigan)
- CCNA Cybersecurity – Cisco

PROJECTS

Real-Time Accident Severity Prediction – Python, Scikit-learn, Power BI • Built AI model (TensorFlow, Scikit-learn) predicting crashes with 85% accuracy on 1M+ records. Deployed scalable APIs on AWS for real-time risk scoring. Created React.js & Power BI dashboards to visualize hotspots and guide safety decisions.	Feb 2024 - May 2024
AI Chatbot for Crash Typing – Python, LLM, Flask, OpenAI API • Built an AI-powered chatbot with OpenAI LLMs to classify and document traffic crash reports. Applied custom prompt tuning & traffic-specific logic, improving classification accuracy. Deployed via Flask API and integrated into internal analytics dashboards for real-time use.	Sep 2024

ADDITIONAL INFORMATION

Publications: "UNVEILING THE EFFICACY: A COMPREHENSIVE ANALYSIS OF ASYMMETRIC ENCRYPTION ALGORITHMS IN DATA TRANSMISSION AND STORAGE SECURITY."